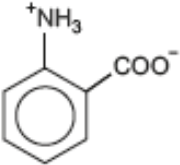


## Section B

| Question |     |       |    | Marking details                                                                                                                                                                                        | Marks available |     |     |       |       |      |
|----------|-----|-------|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |       |    |                                                                                                                                                                                                        | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 9.       | (a) | (i)   |    | sulfur oxide dichloride / thionyl chloride / $\text{SOCl}_2$<br>phosphorus(III) chloride / $\text{PCl}_3$<br>phosphorus(V) chloride $\text{PCl}_5$                                                     | 1               |     |     | 1     |       | 1    |
|          |     | (ii)  |    | it does not absorb in the visible region                                                                                                                                                               | 1               |     |     | 1     |       |      |
|          |     | (iii) |    | loss of $191 - 91 / 100$ (1)<br><br>could be $(\text{C}_2\text{H}_5)_2\text{N}-\text{C}=\text{O}$<br>which has $M_r$ of $29 + 29 + 14 + 28 = 100$ (1)<br><br>accept $\text{C}_5\text{H}_{10}\text{NO}$ |                 |     | 2   | 2     |       |      |
|          | (b) | (i)   |    | alkaline potassium manganate(VII) solution / $\text{MnO}_4^- + \text{OH}^-$                                                                                                                            | 1               |     |     | 1     |       | 1    |
|          |     | (ii)  |    | as the reaction is carried out in alkaline solution the salt of the acid rather than the acid itself is produced                                                                                       |                 |     | 1   | 1     |       | 1    |
|          | (c) | (i)   |    |                                                                                                                      |                 | 1   |     | 1     |       |      |
|          |     | (ii)  |    | the ester contains a basic $-\text{NH}_2$ group which can accept a proton                                                                                                                              |                 | 1   |     | 1     |       |      |
|          |     | (iii) | I  | water is produced as during the esterification reaction / an aqueous solution of sodium carbonate has been added                                                                                       |                 | 1   |     | 1     |       | 1    |
|          |     |       | II | electrically heated / hot water bath                                                                                                                                                                   |                 |     | 1   | 1     |       | 1    |

| Question |     |      |  | Marking details                                                                                                                                                                                                                               | Marks available |     |     |       |       |      |
|----------|-----|------|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |      |  |                                                                                                                                                                                                                                               | AO1             | AO2 | AO3 | Total | Maths | Prac |
|          | (d) | (i)  |  | $2.0 \times 10^{-5}$                                                                                                                                                                                                                          |                 | 1   |     | 1     | 1     |      |
|          |     | (ii) |  | as BA is produced it dissolves preferentially in the hexane (1)<br><br>this removes BA from the reaction and moves the position of equilibrium to the right (1)<br><br>2-aminobenzoic acid remains largely in the aqueous alcohol mixture (1) |                 |     | 3   | 3     |       | 3    |
|          |     |      |  | Question 9 total                                                                                                                                                                                                                              | 3               | 4   | 7   | 14    | 1     | 8    |